How Does Pitch Register Affect Instrument-Family Recognition?

A Controlled Comparison of Handcrafted Acoustic Features and Frozen MERT Embeddings

Full Name: Yuntong Li

Student ID (SID): 530296263

GitHub Username: lytbxqmz111

GitHub Project Repository:
`https://github.com/lytbxqmz111/elec5305-project-530296263`

GitHub Project Site:
`https://lytbxqmz111.github.io/elec5305-project-530296263/`

Video Presentation: `https://drive.google.com/file/d/1t7cOz5LowBdHA8zApOOzeJqaPwMSTwDO/view?usp=sharing`

Abstract

Instrument-family classifiers may exploit the pitch ranges associated with particular instruments rather than relying only on timbral cues. This project investigates how pitch register affects recognition of previously unseen instrument instances in NSynth. We compare a linear classifier trained on 76 handcrafted acoustic features with one trained on mean-pooled embeddings from frozen MERT-v0-public. Using the official instrument-disjoint splits, we select six families with a common MIDI range of 43–69 and construct a test set balanced by family, register, and velocity. We evaluate ordinary and balanced family classification, probe the representations for coarse pitch information, and examine matched sound pairs. We also train separate classifiers in the lower and upper registers and test transfer to the opposite register. On the ordinary test set, handcrafted features and MERT achieve macro-F1 scores of 0.682 and 0.698; on the balanced test set, 0.671 and 0.706. The ordinary test set already shows only a weak family–register association (Cramer's $V = 0.063$), and paired MERT-minus-handcrafted intervals include zero in both conditions. In the cross-register experiment, same-register training outperforms opposite-register training on the same lower-register test notes by 0.20 (handcrafted) and 0.34 (MERT) macro-F1, and on upper-register notes by 0.08 and 0.31; only the handcrafted upper-register interval includes zero. The findings show that stable performance under distribution balancing does not imply strong transfer when training is restricted to one register. The small controlled test sets and differing notes across registers limit causal interpretation.

Contents

1 Introduction **1**

2 Literature Review **1**
2.1 Timbre and pitch are coupled in musical sounds 1
2.2 Interpretable acoustic descriptors as a baseline 2
2.3 Learned representations and disentanglement 2
2.4 Dataset design and the remaining evaluation question 3

3 Methods **3**
3.1 Dataset and controlled subset 3
3.2 Audio representations and family classifiers 4
3.3 Evaluation conditions and metrics 4

4 Results **5**
4.1 Pitch distributions and the controlled test set 5
4.2 Ordinary and balanced family recognition 6
4.3 Pitch decodability and cross-register transfer 7
4.4 Acoustic examples 7

5 Discussion **8**
5.1 What the balanced evaluation establishes 8
5.2 Why cross-register transfer gives a different answer 9
5.3 Pitch information and acoustic interpretation 9
5.4 Limitations and next experiment 10

6 Conclusion **10**

1 Introduction

Recognising a musical instrument from a short note is often described as a timbre-recognition task. In a recording, however, instrument identity is accompanied by other information. Pitch changes harmonic spacing and can alter spectral balance and attack; playing intensity can change both level and tone. Perceptual evidence likewise shows that descriptions of an instrument's timbre vary with register and pitch height [2]. A classifier that predicts an instrument family correctly may therefore use both family-specific acoustic structure and the pitch ranges in which that family commonly occurs. The distinction matters when the aim is to recognise a previously unseen instrument, rather than another note from an instrument already present in training.

NSynth provides a useful setting for examining this issue. It contains isolated, labelled musical notes with metadata for instrument family, source instrument, pitch, and velocity [1]. Its official training, validation, and test splits allow evaluation on source instruments absent from training. Nevertheless, an instrument-disjoint split alone does not equalise the pitch distributions of the families being compared. If those distributions differ, ordinary test accuracy cannot by itself show whether a system has learned cues that remain useful across registers.

The MERT study evaluates NSynth instrument-family and pitch classification as separate downstream tasks [3]. Our study asks a complementary question: how do fixed family classifiers behave after register and velocity are balanced, and when classifier training is restricted to one register?

This report asks: *When classifying previously unseen instrument instances, how strongly does instrument-family recognition depend on pitch register, and do frozen pretrained music representations preserve family identity across pitch more effectively than handcrafted acoustic features?* We study six NSynth families within a shared MIDI range. First, we compare a handcrafted-feature system and a frozen-MERT system on an ordinary test set and on a family-, register-, and velocity-balanced test set, using the same fitted classifiers in both conditions. We then examine performance by register and probe how readily coarse pitch can be decoded from each representation. Finally, a separate experiment trains new classifiers in either the lower or upper register and tests them in both the source and opposite register. Matched sound-pair analyses connect these model results to spectral and temporal changes in the notes themselves.

We reproduce conventional NSynth family classification with handcrafted features and frozen MERT embeddings followed by linear classifiers, together with a simple pitch probe. Our modification is to add a register- and velocity-balanced evaluation set and a separate cross-register transfer experiment. These changes test whether strong ordinary scores persist under controlled distributions and whether a decision rule learned in one register transfers to another. The report next reviews the relevant perceptual and computational literature, describes the controlled methods, presents the results, and discusses their implications and limits.

2 Literature Review

2.1 Timbre and pitch are coupled in musical sounds

Timbre is commonly treated as a multidimensional perceptual quality related to spectral, temporal, and spectrotemporal properties. McAdams emphasises that these properties also interact with fundamental frequency and playing effort [4]. Reymore and colleagues provide direct perceptual evidence: listeners' semantic descriptions of notes from the same instrument changed systematically across registers and pitch heights [2]. Handel and Erickson found that untrained listeners struggled to judge whether notes separated by an octave or more came from the same wind instrument [11]. However, Steele and Williams found that trained musicians retained substantially better performance across larger separations, so one octave should not be treated as a universal boundary [12]. These perceptual results motivate, but do not predict, a machine-classification experiment. They lead us to report performance by register and to test transfer

across widely separated registers, while avoiding a claim that the human result sets a fixed machine threshold.

2.2 Interpretable acoustic descriptors as a baseline

Classical approaches describe a sound through measurements of its short-time spectrum and amplitude envelope. Eronen and Klapuri combined cepstral coefficients with spectral and temporal features for recognition of isolated orchestral notes across instrument pitch ranges [6]. Their study demonstrates the value of combining feature groups for instrument-family recognition, although its dataset and evaluation protocol differ from NSynth and its accuracy cannot be directly compared here. The Timbre Toolbox later organised descriptors derived from Fourier, harmonic, auditory, and energy-envelope representations into spectral, temporal, spectrotemporal, and energetic properties [5]. Together, these studies support a multidimensional acoustic baseline rather than a single "best" timbre measure. MFCCs compactly summarise a Mel-scaled spectral shape; centroid, bandwidth, roll-off, and flatness capture complementary aspects of spectral distribution; and amplitude-envelope and attack descriptors describe how a note develops over time. Their physical meanings make individual classification errors easier to investigate.

Interpretability does not imply pitch invariance. If the fundamental frequency rises, harmonics occupy different frequencies and Mel bands; MFCCs and centroid may consequently change even if the source instrument is unchanged. Roll-off and bandwidth can shift for related reasons. Conversely, an attack-time measure might remain relatively stable for one instrument but vary with register or velocity for another. These are physical and measurement-level reasons to inspect matched notes, rather than treating every feature difference as a difference in instrument identity. The handcrafted system is therefore an informative reference with modest modelling complexity, but its fixed descriptors cannot guarantee that family and register are separated. Peeters and colleagues' multidimensional account informs our combination of spectral and temporal features and the matched-note acoustic analysis [5].

2.3 Learned representations and disentanglement

Pretrained models learn representations from audio rather than using a fixed list of descriptors. In self-supervised learning, training targets are derived from the audio itself, allowing an encoder to learn from recordings without a family label for every training segment. Li and colleagues introduced MERT using this approach and evaluated it on multiple downstream music tasks, including separate NSynth instrument-family and pitch classification tasks [3]. Their results show that MERT embeddings contain information useful for both tasks, but do not establish whether a family decision rule remains reliable across registers.

For the present study, MERT is used as a *frozen* encoder: its pretrained parameters remain fixed while a small classifier learns from the resulting embeddings. This makes it possible to examine a pretrained representation without training a new large network. The trade-off is that embedding coordinates are less directly interpretable than acoustic measurements such as centroid or attack time.

Work on pitch–timbre disentanglement addresses a stronger goal. For example, Luo and colleagues proposed separate latent factors for pitch and timbre in isolated instrument sounds [7]. Such a model explicitly seeks factor separation, whereas MERT is used here as a general-purpose frozen representation; reproducing a disentangling model would add a separate training and validation problem. Lostanlen and Cella instead designed a convolutional architecture for instrument recognition with pitch-related transformations in mind [13]. Their work shows that invariance can be built into a model, whereas our question is empirical: how well do two existing representations support a simple classifier under register shift? This distinction also clarifies the meaning of a *probe*: a simple classifier trained to predict an attribute from fixed embeddings

measures how readily that attribute can be decoded. High pitch-probe performance does not show that a separately trained family classifier uses pitch in its decisions. Equally, a weak probe only limits what that particular classifier can decode. Following the linear-probe approach of Alain and Bengio, we therefore interpret pitch decodability alongside family recognition and cross-register transfer, not as a stand-alone test of disentanglement [10].

2.4 Dataset design and the remaining evaluation question

NSynth's isolated-note design and pitch and velocity metadata make it suitable for controlled comparisons [1]. Its official split keeps training instruments separate from validation and test instruments [8], avoiding the easier evaluation that would result from placing notes of the same source instrument on both sides of a random note-level split. However, source-instrument separation does not require comparable pitch or velocity distributions across families. This strength and limitation of the dataset determine our use of the official split together with a metadata-selected common pitch interval and balanced test manifest.

3 Methods

The methods proceed from a fixed-classifier comparison on ordinary and register–velocity-balanced test sets to a coarse-pitch probe and separately trained cross-register classifiers. Matched acoustic pairs then help interpret the model results. All evaluations retain the official NSynth split; the designs and limits of each comparison are specified below.

3.1 Dataset and controlled subset

We used the official NSynth train, validation, and test splits, containing 289,205, 12,678, and 4,096 four-second, 16-kHz notes, respectively [1, 8]. The unit of prediction was one note, and the label was its instrument family. Source instruments in training were absent from validation and test. We retained the original split labels throughout. Metadata, rather than model outputs, determined the controlled subset before evaluation.

We first inspected family-by-pitch, family-by-velocity, and source-instrument counts. A family needed at least four distinct instruments in each evaluation split. Candidate intervals spanned 24–60 semitones in steps of three, so each could be divided into three equal register bins. Every split–family–register cell had to contain at least 20 notes from at least two instruments and cover at least 65% of the pitches in its bin. Each split and register also needed at least four pitches supported by every selected family; support meant at least five notes from two instruments at a pitch. Feasible designs were ranked by number of families, interval width, and shared-pitch support before sample count. The resulting six families were bass, brass, guitar, keyboard, organ, and string, with a common MIDI range of 43–69. We defined lower (43–51), middle (52–60), and upper (61–69) registers. These criteria favour a common comparison domain over retaining all 11 NSynth families. The selection rules and fixed item identifiers are available in the project repository.

A frozen balanced manifest contains 3,600 training, 1,884 validation, and 594 test notes. Within each split, the six families have equal counts in every register-by-velocity cell; no source instrument supplies more than half of a cell. Velocity strata were the five NSynth values 25, 50, 75, 100, and 127. Sampling used seed 5305. The test set has 99 notes per family. Matching broad registers and velocity removes the direct family–register and family–velocity frequency differences within this sample, while preserving the official split. It does not match every exact pitch or source characteristic. A stricter exact-pitch-by-velocity manifest was constructed as a sensitivity resource, but its 120-note test set was too small for the primary register-level comparison. The broader manifest therefore trades some residual within-bin pitch variation for enough observations to inspect all three registers.

3.2 Audio representations and family classifiers

The handcrafted system extracts 76 features from each note. Audio is processed at its native 16-kHz sample rate. A 1,024-sample Hann-windowed Fourier analysis with a 512-sample hop yields a spectrum approximately every 32 ms. Forty Mel filters and a discrete cosine transform provide 13 MFCCs; frame-to-frame MFCC differences capture short-term change. The mean and standard deviation of each MFCC and difference coefficient contribute 52 features. Means and standard deviations of spectral centroid, bandwidth, 85%-energy roll-off, flatness, and flux add ten more. RMS summaries, zero-crossing, crest factor, temporal-centroid, and attack measurements complete the vector. We use the time between the first pre-peak RMS frames reaching 10% and 90% of peak RMS as an envelope rise-time proxy. RMS frames are spaced 32 ms apart, so short transients are resolved only coarsely; a long value can also reflect a gradual level drift towards a late peak. These quantities provide complementary, physically interpretable views of the spectrum and envelope [5]. They also explain why the acoustic-pair analysis examines both steady-state spectra and onset envelopes.

The learned representation uses the pretrained `MERT-v0-public` encoder with fixed weights [3]. The frozen MERT-v0-public encoder corresponds to MERT-95M-public in [3]. Its 16-kHz input matches NSynth without resampling. We take its final hidden layer, average the frame embeddings over time, and obtain one 768-dimensional vector per note. Mean pooling makes a fixed-size classifier input, although it discards the explicit time order of the frames; the handcrafted representation retains selected temporal summaries instead. No NSynth labels update the encoder. The same selected-family notes are represented by both systems, with cached features keyed to the official split and note identifier. The model revision and extraction settings are pinned in the repository configuration so that the embeddings can be regenerated consistently.

For each representation, a separate `StandardScaler` and class-balanced linear support-vector classifier (`LinearSVC`) were fitted on official training notes from the six families. Linear support-vector machines offer a simple margin-based decision rule [9]; using the same classifier family limits differences due to downstream model complexity. Each scaler estimates its feature means and variances from training data only, then transforms validation and test features. Class weighting counteracts unequal family counts in the ordinary training set. Regularisation parameter $C \in \{0.01, 0.1, 1.0\}$ was selected by validation macro-F1, breaking ties by accuracy and then smaller C; test data were excluded from fitting and model selection. The ordinary classifiers were trained on all 216,611 selected-family training notes across their full pitch ranges; only the reported ordinary test was restricted to MIDI 43–69. Its reported test condition contains 1,244 notes restricted to the shared MIDI interval, while validation-based selection uses the official selected-family validation split. Validation selected $C = 0.1$ for the handcrafted classifier and $C = 1.0$ for MERT. The classifier was fitted once per representation and reused without refitting for the 594-note balanced test condition. This keeps the representation, decision rule, families, and pitch interval fixed while the test distribution changes.

3.3 Evaluation conditions and metrics

For ordinary and balanced tests we report accuracy and macro-F1; balanced-test macro-F1 and family recall are also shown by register, with a confusion matrix for MERT lower-to-upper transfer. Macro-F1 averages the six family scores $F_{1,k} = 2P_kR_k/(P_k + R_k)$ and thus weights families equally. The ordinary-to-balanced change is descriptive: the balanced notes are a subset of the ordinary test, but no interval was calculated for this contrast.

We used 2,000 family-stratified source-instrument cluster-bootstrap replicates (seed 5305), retaining all test notes from each sampled instrument, to obtain percentile 95% intervals. Ordinary and balanced tests share 42 source instruments; within each condition, both representations use the same resampled notes for paired differences. For Table 2, instruments are resampled

within each target register and family, pairing training conditions on the same notes; each target register has 42 instruments. Training sets were sampled once, so their sampling variance is excluded.

A separate pitch probe asks how linearly accessible register information is in each fixed representation. Linear probes provide a limited but useful test of what a representation makes readily classifiable [10]. Here the label is lower, middle, or upper register, rather than instrument family. A new scaler and linear classifier are trained on 81,379 official training notes from the six families inside MIDI 43–69; C is selected using 3,793 validation notes. The selected values were $C = 0.1$ for handcrafted features and $C = 0.01$ for MERT. Probe macro-F1 is then reported on ordinary and balanced test notes. The probe is trained separately from the family model and does not receive the family label as an input. Its score measures decodability under this particular linear classifier and does not show which features the family classifier actually uses.

To test transfer more directly, we train *new* family classifiers on the balanced manifest, separately using lower-register or upper-register training notes. For each source register and representation, a scaler is fitted on its 1,200 training notes and C is chosen using only validation notes from that source register, following the same three-value search. For both lower- and upper-register training, validation selected $C = 0.1$ for handcrafted features and $C = 0.01$ for MERT. The selected classifier is evaluated on both the same and opposite test register: lower→lower versus lower→upper, and upper→upper versus upper→lower. The same-register result provides a reference for a classifier trained with the same number of source-register examples. Lower and upper test sets contain 186 and 192 notes, respectively; both representations use identical manifest rows. The middle register is not a source or target in this experiment. Because this training set is much smaller and narrower than that of the ordinary classifier, its absolute scores must not be read as a direct comparison with the ordinary baseline. Same- and opposite-register test sets also contain different notes, limiting causal claims about pitch alone.

Finally, four representative pairs from the balanced test set provide an acoustic check on the model-based interpretation. Brass–string and keyboard–organ pairs match exact pitch, velocity, and source type while changing family. Brass and organ low–high pairs hold the source instrument and velocity fixed while changing pitch. Waveforms, magnitude spectra, log-Mel spectrograms, spectral envelopes, and RMS attack envelopes show which spectral and temporal structures change or persist. The matched-pitch pairs can reveal family cues beyond pitch, while the within-source pairs show how much a note's own acoustic structure changes with register. These examples illustrate possible mechanisms and are not population-level estimates.

Pair selection was fixed from metadata in the balanced test manifest, without using classifier predictions. Two pairs match exact MIDI pitch, velocity, and source type while changing family; two hold the source instrument and velocity fixed while moving from the lower to the upper register. These four examples were chosen to illustrate both comparison types, not sampled to estimate their population frequencies. For each note, the Mel spectral envelope is the time-mean power in each of 80 Mel bands. Envelope similarity is the cosine similarity of these non-negative, linear-power vectors, without mean subtraction; decibel-scaled envelopes are used only for display. Attack-envelope correlation is Pearson correlation between the two peak-normalised RMS sequences over their common initial length. The expected fundamental-frequency ratio is calculated from the nominal MIDI pitches, $f(p) = 440\,2^{(p-69)/12}$ Hz; it is not an estimate of the recorded fundamental frequency.

4 Results

4.1 Pitch distributions and the controlled test set

Figure 1 shows the composition of the ordinary six-family test set across the three registers within MIDI 43–69. The association between family and broad register is weak (Cramer's $V = 0.063$, $\chi^2 = 9.90$, $N = 1{,}244$). For example, the upper-register share ranges from 25.6% for

string to 39.4% for brass. The balanced test manifest contains 594 notes, or 99 per family; its lower, middle, and upper registers contain 186, 216, and 192 notes, respectively. Thus balancing removes the remaining family–register frequency differences in this sample, but the ordinary test set did not contain a strong family–register association to begin with.

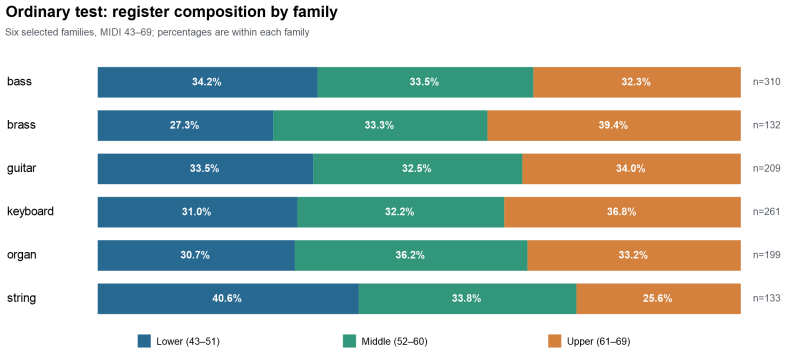


Figure 1: Register composition of the ordinary six-family test set within MIDI 43–69. Each bar sums to 100% within one family; colours indicate lower (43–51), middle (52–60), and upper (61–69) registers.

4.2 Ordinary and balanced family recognition

Table 1 compares the *same fitted classifier* on ordinary and balanced test notes. Handcrafted macro-F1 decreases by 0.0108, from 0.6816 to 0.6707; MERT increases by 0.0073, from 0.6982 to 0.7055. MERT has the higher absolute macro-F1 in both conditions, but the changes associated with balancing are small for both representations. Accuracy rises in the balanced condition for both systems, illustrating why accuracy and macro-F1 should be reported together when class counts change. However, the paired MERT-minus-handcrafted confidence intervals include zero in both conditions. The observed MERT advantage is therefore uncertain at the level of the sampled source instruments.

Table 1: Six-family classification. Macro-F1 values are followed by 95% instrument-cluster bootstrap confidence intervals. The final column is the paired MERT-minus-handcrafted macro-F1 difference within each test condition.

Test condition	Representation	Macro-F1 [95% CI]	Accuracy	MERT − handcrafted [95% CI]
Ordinary	Handcrafted	0.6816 [0.571, 0.773]	0.6359	
	Frozen MERT	0.6982 [0.599, 0.779]	0.6712	0.0166 [−0.052, 0.094]
Balanced	Handcrafted	0.6707 [0.555, 0.765]	0.6650	
	Frozen MERT	0.7055 [0.602, 0.791]	0.6987	0.0348 [−0.041, 0.117]

The balanced aggregate conceals register-specific differences. Handcrafted macro-F1 is 0.6593, 0.6615, and 0.6793 in the lower, middle, and upper registers; MERT obtains 0.7324, 0.7178, and 0.6571. Numerically, MERT leads in the lower and middle bins, while handcrafted features lead slightly in the upper bin; these register-level comparisons are exploratory without intervals. Family recall is also uneven: handcrafted bass recall falls to 0.2778 in the middle register, and MERT bass recall falls from 0.7742 in the lower register to 0.3438 in the upper. These patterns show why one pooled score cannot describe the behaviour of every family across pitch.

Figure 2 displays the complete family-by-register recall pattern for both representations. It shows whether changes in register-level macro-F1 are spread across families or concentrated in particular classes.

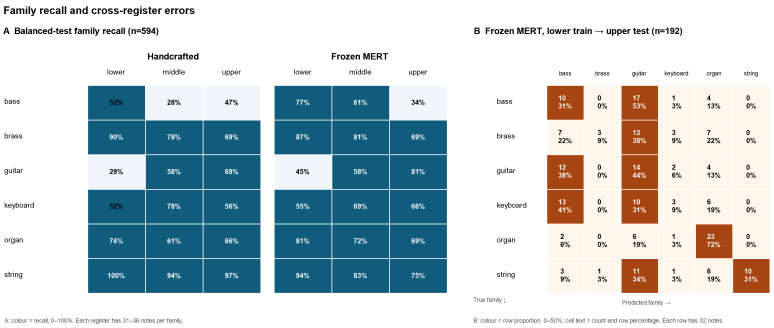


Figure 2: (A) Per-family recall by register on the 594-note balanced test set for handcrafted features and frozen MERT; colour represents recall. (B) Row-normalised confusion matrix for the frozen-MERT classifier trained on lower-register notes and tested on 192 upper-register notes. Rows are true families and columns are predicted families. Colour represents the within-row proportion; cells show counts and row percentages.

4.3 Pitch decodability and cross-register transfer

The coarse-pitch probe predicts register from the two fixed representations. On the balanced test, its macro-F1 is 0.7650 for handcrafted features and 0.9783 for MERT. Corresponding ordinary-test scores are 0.7845 and 0.9856. Register information is therefore highly linearly accessible in MERT embeddings, while MERT also has a numerically higher balanced family macro-F1. Probe performance measures accessible information, not whether the family classifier used that information.

Table 2 compares training registers within each fixed test set. For handcrafted features, same-register training has a macro-F1 advantage of 0.2026 on lower-register test notes but only 0.0786 on upper-register notes, a numerically larger lower-test advantage. The corresponding advantages for MERT are 0.3357 and 0.3117. These within-column comparisons avoid mixing lower- and upper-test-set difficulty. MERT's same-register scores (0.6203 and 0.6235) are also lower than the handcrafted scores (0.7793 and 0.7338) under this restricted-training protocol. The numerically larger MERT advantages, whose between-system differences lack intervals, therefore should not be interpreted as an intrinsic ranking of representation robustness. The bootstrap intervals for MERT's same-register advantages exclude zero in both test registers. For handcrafted features, the interval excludes zero on lower-register test notes but includes zero on upper-register test notes.

The row-normalised confusion matrix in Fig. 2B shows where these errors go. Of the 32 upper-register brass notes, 12 are predicted as guitar and seven each as bass and organ. Of the 32 keyboard notes, 13 are predicted as bass and ten as guitar. These concentrations identify specific family boundaries that fail under lower-to-upper transfer.

4.4 Acoustic examples

The matched sound pairs illustrate acoustic changes relevant to these classification results. At the same MIDI pitch and velocity, the brass note has a spectral centroid of 3,930 Hz versus 1,628 Hz for the string note, and their Mel-envelope cosine similarity is 0.339. A second fixed-pitch comparison, at MIDI 43 and velocity 100, contrasts keyboard and organ notes: their Mel-envelope similarity is 0.198, and the organ's measured 10–90% RMS rise time is 1.600 s longer than the keyboard's, reflecting a slow increase towards its peak under this measurement (Fig. 3). These pairs retain substantial family-related differences when pitch metadata is fixed. In the same-source brass pair, raising pitch from MIDI 43 to 68 multiplies the expected fundamental frequency by 4.24 but the centroid by only 1.26; its attack envelopes remain strongly correlated

Table 2: Cross-register macro-F1 by training and test register. Within each test column, the same-register training advantage is the score from training in that register minus the score from training in the opposite register. Lower and upper test sets contain 186 and 192 notes, respectively. Intervals for the same-register advantages use paired, family-stratified source-instrument cluster bootstrapping.

Representation	Train register	Test lower	Test upper
Handcrafted	Lower	**0.7793**	0.6552
Handcrafted	Upper	0.5767	**0.7338**
Same-register advantage [95% CI]		0.2026 [0.101, 0.309]	0.0786 [−0.020, 0.177]
Frozen MERT	Lower	**0.6203**	0.3118
Frozen MERT	Upper	0.2846	**0.6235**
Same-register advantage [95% CI]		0.3357 [0.214, 0.447]	0.3117 [0.231, 0.390]

(0.906). For the organ pair, the upper note's attack is 0.384 s longer than the lower note's. These examples show both persistent cues and within-instrument changes across register. Four selected pairs cannot establish how often each pattern occurs in the full dataset; the other pair plots and measurements are available in the project repository.

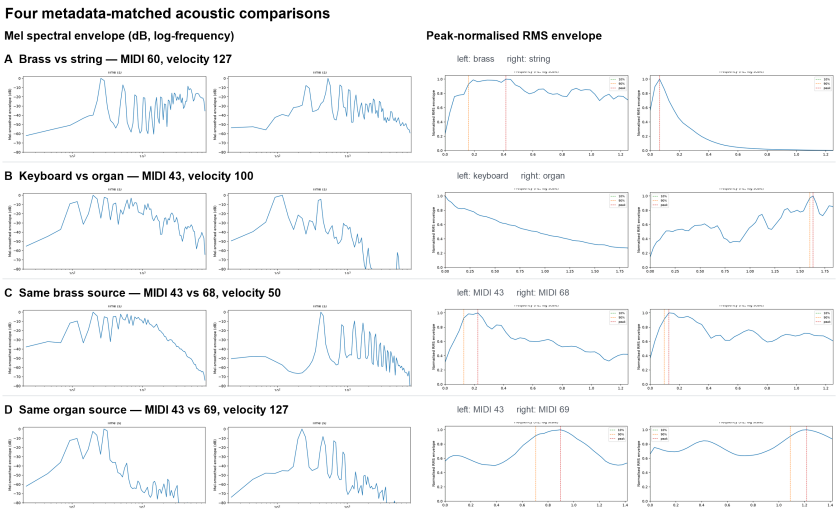


Figure 3: Four metadata-matched note pairs. Each row compares two notes using their time-averaged Mel spectral envelopes (left panel, displayed in dB) and peak-normalised RMS envelopes (right panel). Rows A and B hold MIDI pitch and velocity fixed while changing instrument family; rows C and D hold source instrument and velocity fixed while changing from a lower to an upper pitch. Within each panel, the two notes appear on the left and right, respectively.

5 Discussion

5.1 What the balanced evaluation establishes

The ordinary test set already has only a weak family–register association, so balancing removes a small distributional difference in this sample. The aggregate changes in Table 1 are small but lack intervals for the ordinary-to-balanced contrast; they say little about a stronger confound. MERT's numerically higher scores do not establish superior pitch robustness.

This finding has a precise boundary. The comparison changes which test notes are included; it does not remove pitch information from the audio or the representations. The bins also match

broad register rather than exact MIDI pitch. A classifier trained on many pitches may learn different useful cues in different regions and still score consistently after balancing. Moreover, the stable aggregate hides local variation: Exploratorily, MERT's macro-F1 is 0.7324 in the lower register and 0.6571 in the upper, while handcrafted scores vary less; these between-register differences have no intervals. The family-level bass results show that even this register-level summary can conceal a pronounced weakness for one class. The balanced test therefore supports a conclusion about the overall evaluation distribution, not a claim that either representation is invariant to pitch.

5.2 Why cross-register transfer gives a different answer

The cross-register experiment changes the learning problem: each classifier sees only 1,200 training notes from one register. Within each fixed test set, Table 2 supports a same-register advantage for MERT in both targets and for handcrafted features on lower notes; the handcrafted upper-target interval includes zero. The handcrafted advantage is numerically larger on lower than upper notes, without an interval for that difference.

The lower and upper test regions are separated by 10–26 semitones, depending on the particular notes compared. This range overlaps the pitch separations discussed in perceptual studies of instrument recognition [11, 12]. Those studies concern human listeners, whereas our experiment evaluates classifiers trained and tested on different note samples. Their findings motivate examining wide register shifts but do not provide a one-octave threshold for machine performance.

MERT's numerically larger same-register advantages do not establish intrinsically weaker pitch robustness: its restricted-training same-register scores are already lower than the handcrafted scores, and the full-range classifiers used far more training notes. In lower-to-upper transfer, MERT identifies only 3 of 32 brass and 3 of 32 keyboard notes, versus 15 of 32 for each with handcrafted features. Limited training data relative to its 768-dimensional embeddings is one possible explanation requiring a separate diagnostic; test-note and source-instrument composition may also matter.

The published MERT-95M-public NSynth instrument result is 70.4% accuracy over 11 families using an MLP probe [3]. Our six-family, MIDI 43–69 linear-SVM protocol differs, so its accuracy is not directly comparable with ours; our tests instead examine cross-register behaviour of a frozen encoder's downstream classifier.

5.3 Pitch information and acoustic interpretation

The pitch probe provides a complementary measurement. Coarse register is almost perfectly linearly decodable from MERT on the balanced test (macro-F1 0.9783), while MERT family classification is numerically higher than the handcrafted baseline in that same test. These measurements show that MERT retains accessible information about both pitch and family. A probe establishes what a specified classifier *can decode*, not which coordinates or cues the independently trained family classifier *actually uses*. It is therefore consistent with both the MERT benchmark's separate pitch and instrument tasks [3] and the literature treating timbre and pitch as interacting properties of musical notes [4, 2].

The acoustic pairs make this interaction concrete. At fixed pitch and velocity, brass and string differ substantially in spectral centroid, while keyboard and organ differ in their measured RMS rise profiles. Family cues therefore remain when these two metadata variables are held constant. Within one brass source, however, a 25-semitone pitch increase multiplies the expected fundamental frequency by 4.24 but the centroid by only 1.26. In the organ pair, attack duration also changes across register. These observations are compatible with a classifier encountering different feature patterns for the same family at different pitches. They do not show that the selected acoustic changes caused the observed classification errors, and four chosen pairs cannot represent the full family distributions.

5.4 Limitations and next experiment

The chief limitation is the size and structure of the controlled test data. The balanced test contains 594 notes, and each opposite-register comparison uses only 186 or 192 target notes. All balanced test notes occur in the ordinary test set, while lower and upper test notes differ. Exact pitch, source-instrument composition, and other NSynth qualities may therefore contribute to a score difference despite matching broad register and velocity. The official split protects against overlap between training instruments and evaluation instruments. Instrument-cluster bootstrap intervals quantify sampling uncertainty, but the modest number of source instruments makes them broad. In particular, the paired intervals do not establish a reliable MERT advantage over handcrafted features in either the ordinary or balanced test. Bootstrapping also cannot eliminate differences in exact pitch, notes, or source-instrument composition between conditions. The strict exact-pitch test manifest contains only 120 notes and cannot solve this precision problem on its own.

The next iteration should strengthen the controlled comparison before expanding the model collection. A larger test set with more source instruments and closer matching of exact pitches would reduce residual differences and improve precision. Repeating the cross-register design with more training examples per source register and several register pairs would test whether its loss decreases with coverage. Finally, comparing MERT layers or temporal pooling strategies under the same frozen protocol would examine whether the present final-layer mean embedding is responsible for part of the transfer gap. NSynth's isolated sample-library notes also limit generalisation to polyphonic or differently recorded music; an external dataset would address that later, once the within-NSynth finding is better established.

6 Conclusion

This project asked whether instrument-family recognition on previously unseen NSynth instruments depends on pitch register, and whether frozen MERT embeddings preserve family identity across pitch more effectively than handcrafted acoustic features. In the full-range classification experiment, MERT obtained a numerically higher macro-F1 on both ordinary (0.6982 versus 0.6816) and register- and velocity-balanced (0.7055 versus 0.6707) tests. Paired instrument-cluster confidence intervals include zero for the MERT-minus-handcrafted difference in both main test conditions. Balancing changed either system's overall macro-F1 by less than 0.011 descriptively, giving little evidence of a large aggregate effect from the family–register frequency differences controlled here. However, same-register training had a supported advantage for MERT in both test registers and for handcrafted features on lower-register notes; MERT's advantages were numerically larger, without a between-system interval. Its same-register scores were also lower under this restricted-training protocol, so the larger advantage does not by itself establish that the MERT representation is inherently less pitch-robust. High pitch decodability from MERT and the matched acoustic examples are consistent with pitch and family cues coexisting in the notes and their representations.

The answer is therefore conditional on the evaluation: stable performance after broad distribution balancing does not imply that a classifier learned in one register will transfer well to another. The contribution is a reproducible comparison that makes this distinction visible while retaining the official instrument-disjoint training split. Because the controlled test is small and register conditions contain different notes, the study does not establish a causal measure of pitch invariance or a general ranking of the two representations. A larger, more closely matched test with more source instruments is the clearest next step.

References

[1] J. Engel, C. Resnick, A. Roberts, S. Dieleman, D. Eck, K. Simonyan, and M. Norouzi, "Neural audio synthesis of musical notes with WaveNet autoencoders," in *Proceedings of the 34th International Conference on Machine Learning*, 2017, pp. 1068–1077.

[2] L. Reymore, J. Noble, C. Saitis, C. Traube, and Z. Wallmark, "Timbre semantic associations vary both between and within instruments: An empirical study incorporating register and pitch height," *Music Perception*, vol. 40, no. 3, pp. 253–274, 2023.

[3] Y. Li *et al.*, "MERT: Acoustic music understanding model with large-scale self-supervised training," in *International Conference on Learning Representations*, 2024.

[4] S. McAdams, "The perceptual representation of timbre," in *Timbre: Acoustics, Perception, and Cognition*, K. Siedenburg, C. Saitis, S. McAdams, A. N. Popper, and R. R. Fay, Eds. Springer, 2019, pp. 23–57.

[5] G. Peeters, B. L. Giordano, P. Susini, N. Misdariis, and S. McAdams, "The Timbre Toolbox: Extracting audio descriptors from musical signals," *Journal of the Acoustical Society of America*, vol. 130, no. 5, pp. 2902–2916, 2011.

[6] A. Eronen and A. Klapuri, "Musical instrument recognition using cepstral coefficients and temporal features," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*, vol. 2, 2000, pp. II-753–II-756, doi: 10.1109/ICASSP.2000.859069.

[7] Y.-J. Luo, K. W. Cheuk, T. Nakano, M. Goto, and D. Herremans, "Unsupervised disentanglement of pitch and timbre for isolated musical instrument sounds," in *Proceedings of the International Society for Music Information Retrieval Conference*, 2020, pp. 700–707.

[8] TensorFlow Datasets, "NSynth dataset documentation," `https://www.tensorflow.org/datasets/catalog/nsynth`, accessed Nov. 2026.

[9] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, pp. 273–297, 1995.

[10] G. Alain and Y. Bengio, "Understanding intermediate layers using linear classifier probes," arXiv:1610.01644, 2016.

[11] S. Handel and M. L. Erickson, "A rule of thumb: The bandwidth for timbre invariance is one octave," *Music Perception*, vol. 19, no. 1, pp. 121–126, 2001, doi: 10.1525/mp.2001.19.1.121.

[12] K. M. Steele and A. K. Williams, "Is the bandwidth for timbre invariance only one octave?," *Music Perception*, vol. 23, no. 3, pp. 215–220, 2006, doi: 10.1525/mp.2006.23.3.215.

[13] V. Lostanlen and C.-E. Cella, "Deep convolutional networks on the pitch spiral for musical instrument recognition," in *Proceedings of the International Society for Music Information Retrieval Conference*, 2016.